

How to make the diagnosis

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Wherever the art of medicine is loved, there is also a love of humanity.

Hippocrates (c.460-370 BCE)

This chapter should be read in conjunction with Chapter 14. These chapters are very different from their fourth edition predecessors. In earlier editions of *Clinical Sports Medicine*, we argued for the importance of accurate pathological diagnosis. At times, this is realistic. A player falls on his or her wrist and sustains an acute scaphoid fracture. You can make a clinical diagnosis and confirm it with imaging. So far, so good. Accurate tissue diagnosis remains a foundation *where appropriate*.

In this fifth edition, we recognise that we previously overestimated the proportion of settings where *accurate, very specific* tissue diagnosis is possible. Consider patients who present with low back pain. Is the tissue diagnosis the facet joint, vertebra, fascia, muscle or disc? In these cases, can investigation, such as magnetic resonance imaging (MRI) provide the holy grail of tissue diagnosis? No, MRI cannot.

This chapter reinforces Chapter 14's key message—the art of diagnosis takes place against a backdrop of patient-based probabilities. Honest teachers will explain that the clinician is narrowing the odds from the 'pre-test probability' to a new 'post-test probability'. (See Likelihood ratios and Fagan's Nomogram, Chapter 14.)

DOES 'DIAGNOSIS' MEAN 'TISSUE DIAGNOSIS'?

Is there a difference between a diagnosis of: 1. 'swimmer's shoulder'; 2. 'rotator cuff tendinopathy'; and 3. 'shoulder pain related to training load errors and suboptimal biomechanics'? The first of these ('swimmer's shoulder') represents a 1970s advance in sports medicine. Alongside labels such as tennis elbow, goalie's elbow, hockey player's groin and footballer's ankle, 'diagnosis' consisted

of the name of a sport paired with a commonly injured body part. This was an advance at the time, as many doctors were not aware of sports medicine conditions and sports physiotherapy had not yet emerged in many countries.

The major limitation of this type of label is lack of precision. Each of those labels (e.g. hockey groin) could apply to a number of pathological entities which may benefit from distinct treatment. Such labels have no place in 21st-century sports medicine.

What about 'rotator cuff tendinopathy'? Is that a valid diagnosis? The fourth edition of *Clinical Sports Medicine* asserted that accurate pathological diagnosis was essential for the following reasons.

- *It enabled the clinician to explain the problem and the natural history of the condition to the athlete, who will want to know precisely for how long he or she will be affected. A patient may present with an acute knee injury but the diagnosis of anterior cruciate ligament (ACL) tear has markedly different implications to the diagnosis of minor meniscal injury.*
- *It enabled optimum treatment. Numerous conditions have similar presentations but markedly different treatments. For example, consider the differences in treatment between: lateral ligament sprain of the ankle and osteochondral fracture of the talus; patellofemoral joint syndrome and meniscal tear; and hamstring tear and hamstring pain referred from the lumbar spine.*
- *It enabled optimum rehabilitation prescription. For example, rehabilitation after shin pain due to stress fracture will be more gradual than that after identical shin pain due to chronic compartment syndrome.*

In that edition, we acknowledged that ‘occasionally’ it can be impossible to make a precise pathological diagnosis as in the case of back pain mentioned above. In such cases, it is still possible to exclude certain causes of low back pain (e.g. spondylolysis) and identify abnormalities such as areas of focal tenderness, altered soft tissue consistency or abnormalities of range of motion. Treatment then aims to resolve the condition. How treatment affects symptoms and signs can help determine how each particular abnormality contributes to the overall picture. An important final step in management is to address the cause/s: intrinsic and extrinsic.

PRACTICE PEARL

Our current perspective is that we overestimated the number of cases where a ‘tissue diagnosis’ is accurate and we certainly underestimated the role of the psychosocial factors in the patient’s perception of the injury.

The biopsychosocial model featured in the fourth edition of *Clinical Sports Medicine* but only in the chapter on neck pain, rather than in Part A: Fundamental Principles. We failed to feature the biopsychosocial model in the diagnosis or treatment chapters but amend that in this edition (Fig. 15.1).

It is widely appreciated that in many clinical settings ‘accurate tissue diagnosis’ is impossible. Despite that, the clinician can guide the patient by assessing impairments, documenting the patient’s level of function, providing reassurance based on epidemiological evidence of clinical outcomes and natural history. In sum, the clinician contributes his/her knowledge of best evidence rehabilitation to the shared decision.^{1,2}

KEYS TO ACCURATE DIAGNOSIS

Diagnosis relies on taking a careful history, performing a thorough physical examination and using appropriate investigations. Today, patient-reported outcome measures (PROMs) may also be essential as part of the initial assessment and this is introduced in the next chapter (Chapter 16). Useful PROMs are also included in the regional injury chapters in Part B of this book. ‘Over-reliance on sophisticated investigations and neglected clinical skills’ has been a mantra of senior medical educators for decades and the problem may be worsening.

Keys to accurate diagnosis in patients presenting with apparent musculoskeletal pain include:

- the age of the patient
- the mechanism of injury

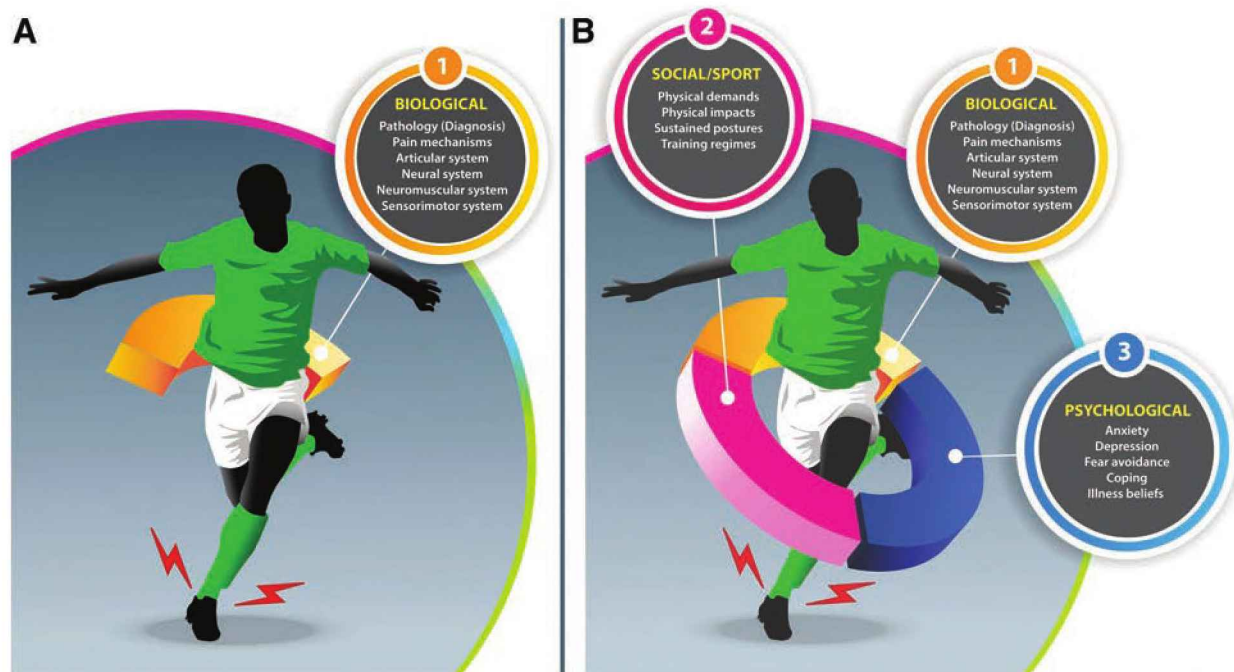


Figure 15.1 The biopsychosocial (BPS) model of disease provides an important context for the patient’s experience (a) In a purely biomedical model, pathology explains the patient’s pain experience and this is considered virtually equivalent to the patient’s experience (b) The biopsychosocial model provides a holistic approach to diagnosis. Many factors, not only the tissue pathology, should be taken into account. Psychological and social/sport factors, depicted here, contribute to a more complete diagnosis. See also Chapters 5 and 6 that explain pain in sports injuries

- whether the symptoms are of musculoskeletal origin (Chapter 7)
- possible local causes of the patient's symptoms
- sites that could be referring pain to the site of the symptoms (Chapters 5, 6)
- the relevant kinetic chain (e.g. the back and lower limb in a shoulder injury of a tennis player)
- biomechanics (Chapters 8 and 9)
- other possible causative factors (e.g. metabolic)
- patient expectations and understanding.

HISTORY

History remains the keystone of accurate diagnosis; it will provide the diagnosis in the majority of cases. Please consider the following principles when taking a history.

Allow enough time

The patient must feel that the clinician has time available to allow the story to unfold, otherwise important symptoms will not surface. In addition to the details of the injury, there must be time to take the history of the training program including details of training workload where this is available. Look into possible causes of injury. Diet history may be appropriate (i.e. contributing to stress fracture likelihood). As a minimum, 30 minutes is required to assess a patient with a new injury but in complex chronic cases up to one hour may be necessary.

Be a good listener

The clinician must let the story unravel. Appropriate body language and focus on the patient, and not the medical record or computer screen, will aid this. The sports clinician is in the fortunate position that many patients have good body awareness and are generally able to describe symptoms very well.

Know the sport

It is helpful to understand the technical demands of a sport when seeing an athlete as this engenders patient confidence. More importantly, knowledge of the biomechanics and techniques of a particular sport (see Chapter 9) can assist greatly in both making the primary diagnosis and uncovering the predisposing factors.

Discover the exact circumstances of the injury

The first task in history taking is to determine the exact context of the injury. The patient's age is the first clue: knee pain in an adolescent is more likely to be an apophysitis such as Osgood-Schlatter disease while in an elderly person degenerative causes are more likely; 'tennis elbow' in a child would be unusual but not in

middle age. Next, the mechanism of injury warrants careful consideration. Most patients will be able to describe in considerable detail the mechanism of injury. In acute injuries, this is the single most important clue to diagnosis. For example, an inversion injury to the ankle strongly suggests a lateral ligament injury, a valgus strain to the knee may cause a medial collateral ligament injury, and a pivoting injury accompanied by a 'pop' in the knee and followed by rapid swelling suggests an ACL injury.

Obtain an accurate description of symptoms, both at the time of injury and at the initial consultation

Common musculoskeletal symptoms include pain, swelling, instability and loss of function.

Pain

Consider the characteristics of the patient's pain.

1. Location. Note the exact location of pain. Detailed knowledge of surface anatomy can be invaluable. If the pain is poorly localised or varies from site to site, consider the possibility of referred pain.
2. Onset. Speed of onset helps determine whether the pain is due to an acute or overuse injury. Was the onset of pain associated with a snap, crack, tear or other sensation?
3. Severity. Severity may be classified as mild, moderate or severe. Using a pain score (e.g. numerical rating scale [NRS]) or a specific outcome measure may be useful to gauge severity and monitor improvement. Another very helpful method is to have the patient detail the most important activity for them, and rate that with a score. Assess the severity of the pain immediately after the injury and subsequently. Was the patient able to continue the activity?
4. Irritability. This refers to the level of activity required to provoke pain and how long it subsequently takes to settle. The degree of irritability is especially important because it affects how vigorously the examination should be performed and how aggressive the treatment should be.
5. Nature. This refers to the quality of the pain. It is important to allow patients to describe pain in their own words. For example, 'burning pain' can suggest neural involvement.
6. Behaviour. Is the pain constant or intermittent? What is the time course of the pain? Is it worse on waking up or does it worsen during the day? Does it wake the patient at night?
7. Pain site and distribution. Does the pain radiate at all? If so, where?

8. Aggravating factors. Which activity or posture aggravates the pain?
9. Relieving factors. Is the pain relieved by rest or the adoption of certain postures? Do certain activities relieve the pain? Is the pain affected by climatic changes (e.g. cold weather)?
10. Associated features. These include swelling, instability, sensory symptoms (such as pins and needles, tingling or numbness) and motor symptoms (such as muscle weakness).
11. Previous treatment. What was the initial treatment of the injury? Was ice applied? Was firm compression applied? Was the injured part immobilised? If so, for how long? What treatment and rehabilitation has been performed and what effect did that treatment have on the pain? The latter is particularly important when you are providing a second or subsequent opinion in cases of long-standing problems.

Swelling

Immediate swelling following an injury may indicate a severe injury such as a fracture or major ligament tear accompanied by haemarthrosis. Record the degree of swelling (mild, moderate or severe) and subsequent changes in the amount of swelling.

Instability

Any history of giving way or feeling of instability is significant. Try to elicit the exact activity that causes this feeling. For example, in throwing, does the feeling of instability occur in the cocking phase or the follow-through?

Function

It is important to know whether the athlete was: 1. able to continue activity without any problems immediately after the injury happened; 2. able to continue with some restriction; or 3. unable to continue. Note subsequent changes in function with time. When patients present with chronic conditions, or have been receiving treatment already, see what level of function they have reached. Have they just failed to rehabilitate to the necessary level of function?³ (See Chapter 18.)

History of a previous similar injury

If the athlete has had a previous similar injury, record full details of all treatment given, response to each type of treatment and whether any maintenance treatment or exercises have been performed following initial rehabilitation. Previous injury is the most important risk factor for recurrence.

Other injuries

Past injuries may have contributed to the current injury; for example, an inadequately rehabilitated muscle tear that has led to muscle imbalance and a subsequent overuse injury. Because of the importance of spinal abnormalities as a potential component of the athlete's pain (Chapter 5), the patient should always be questioned about spinal symptoms, especially pain and stiffness in the lower back or neck. Past or present injuries in body parts that may at first seem unrelated to the present injury may also be important. For example, a hamstring injury in a throwing athlete can impair the kinetic chain leading to the shoulder, alter throwing biomechanics and thus contribute to a rotator cuff injury.

General health

Is the patient otherwise healthy? Musculoskeletal symptoms are not always activity related (Chapter 7). Red flags for serious medical conditions masquerading as sports injuries include:

- no specific mechanism of injury
- pain unrelated to activity
- night pain
- associated symptoms and signs such as weight loss, fever, malaise, lymphadenopathy
- poor response to treatment.

Consider asking specifically about diabetes, possible undiagnosed inflammatory conditions (e.g. skin problems of psoriasis, bowel complaints associated with spondyloarthropathies) and familial conditions.

Work and leisure activities

Work and leisure activities can play a role in both the aetiology and subsequent management of an injury. For example, a patient whose job involves continual bending or who enjoys gardening may aggravate his or her low back pain. It is important to know about these activities and to ascertain whether they can be curtailed.

Consider why the problem has occurred

Predisposing factors should be considered not only in overuse injuries, but also in medical conditions and in acute injuries. In an athlete struggling to return to play after an ACL injury for example, there may be an underlying psychological component.

An athlete with an acute hamstring tear may have a history of low back problems or, alternatively, a history of a previous inadequately rehabilitated tear. Recurrence can only be prevented by eliminating the underlying cause.

Training/activity history

In any overuse injury, a comprehensive training history is required. This is best done as a weekly diary as most athletes train on a weekly cycle (Chapter 12). It should contain both the quantity and quality of training and describe any recent changes. Note the total amount of training (distance or hours depending on the sport) and training surfaces. Ideally, the patient will have a detailed training history that includes perception of how hard training was. Training can be measured as (i) 'internal load', such as measured by rating of perceived exertion (RPE), or (ii) 'external load', such as training volume in distance covered or balls pitched or bowled. Continual activity on hard surfaces or a recent change in surface may predispose to injury. In running sports, pay particular attention to footwear (Chapter 8). For both training and competition shoes, note the shoe type, age and wear pattern. Record recovery activities such as soft tissue therapy, spa/sauna and hours of sleep. Asking patients to bring a training diary or history to the consultation is a useful pre-emptive tactic.

Equipment

Inappropriate equipment may predispose to injury (Chapter 12). For example, a bicycle seat that is set too low may contribute to patellofemoral pain.

Technique

Patients should discuss technique problems that either they, or their coach, have noted. Faulty technique may contribute to injury. For example, a 'wristy' backhand drive may contribute to extensor tendinopathy at the elbow (Chapter 9).

Overtraining

Symptoms such as excessive fatigue, recurrent illness, reduced motivation, persistent soreness and stiffness may point to overtraining as an aetiological factor.

Psychological factors

Injury can be caused or exacerbated by a number of psychological factors that may relate to sport (i.e. pressure of impending competition), or may concern personal or business life. The clinician needs to consider this possibility and approach it sensitively.

Nutritional factors

Inadequate nutrition can predispose to the overtraining syndrome and may play a role in the development of musculoskeletal injuries. Eating disorders may also inhibit

healing. In an athlete presenting with excessive tiredness, a full dietary history is essential.

Drugs: prescription and others

As with all complete medical histories, a history of medication use may be appropriate. Fluoroquinolone antibiotics may be the cause of a patient's tendinopathy, while cold and influenza medications have been associated with arrhythmias. Athletes may take social drugs such as cocaine, and performance-enhancing anabolic steroids have a range of systemic side effects. Athletes are unlikely to volunteer this information, and direct questioning is not only appropriate but reveals a good clinical understanding of the context of injury in sport.

History of exercise-induced anaphylaxis

Exercise-induced anaphylaxis (EIA) and food-dependent, exercise-induced anaphylaxis (FDEIA) are rare, but potentially life-threatening clinical syndromes in which association with exercise is crucial. It is a clinical syndrome in which anaphylaxis occurs in conjunction with exercise. Given the rarity of the condition, our current understanding relies on case studies alone and is acknowledged as being very limited.⁴

Determine the importance of the sport to the athlete

The level of commitment to the sport, which will not necessarily correlate with the athlete's expertise, has a bearing on management decisions. Be aware of the athlete's short- and long-term future sporting commitments to schedule appropriate treatment and rehabilitation programs.

Differential diagnosis

At the conclusion of taking the history, it is important to consider the differential diagnosis and the possible aetiological factors. Then proceed to a thorough focused examination.

PHYSICAL EXAMINATION

In this practical chapter, we share a number of general principles that might speed the novice clinician towards better physical examination. We acknowledge the limitations of physical examination (Chapter 14)⁵ and of imaging (below), but they still provide valuable information that increases the likelihood of making the correct diagnosis and thus of choosing evidence-based treatment for a patient's complaints.

Develop a routine

Use a specific routine for examining each joint, region or system as this forms a habit and allows you to concentrate on the findings and their significance rather than thinking of what to do next. In Part B we outline a routine for examining each body part.

Remember that in most sports the athlete functions in the upright position. Avoid the temptation to only examine the patient in a supine or prone position. Moreover, running sports involve supporting the body on one leg or the other most of the time. Simulate this in your examination. Consider starting the examination assessing posture and having the patient perform various stress tests on one leg (e.g. hopping, one-legged squat).

Where relevant, examine the other side

With some aspects of the examination (e.g. ligamentous laxity or muscle tightness), it is important to compare sides using the uninjured side as a control.

Consider possible causes of the injury

Try to ascertain the cause of the injury. It is not sufficient to examine the painful area only (e.g. the Achilles tendon). Acknowledge the importance of the kinetic chain. Examine joints, muscles and neural structures proximal and distal to the injured area, seeking predisposing factors (e.g. limited dorsiflexion of the ankle, tight gastrocnemius-soleus complex, lumbar facet joint dysfunction). A useful tip is to examine all the areas that may possibly contribute to the injury before honing in on the site of pain.

Attempt to reproduce the patient's symptoms

It is helpful to reproduce the patient's symptoms if possible. This can be achieved both by active and/or passive movements and by palpation either locally or, in the case of referred pain, at the site of referral. It may require you to send the patient for a run or some other test of function prior to examination (see below).

Assess local tissues

Assess the joints, muscles and neural structures at the site of pain for tenderness, tissue feel and range of motion.

Assess for referred pain

Assess the joints, muscles and neural structures that may refer to the site of pain (Chapter 4).

Assess neural mechanosensitivity

Neural mechanosensitivity (Chapters 5 and 6) should be assessed using one or more of the neurodynamic tests (see below).

Examine the spine

Many injuries have a spinal component to the pain or dysfunction. Abnormal neural mechanosensitivity suggests a possible spinal component. In lower limb injuries, examine the lumbar spine and the thoracolumbar junction. In upper limb injuries, examine the cervical and upper thoracic spines. Examine for hypomobility of isolated spinal segments as this may contribute to distant symptoms. See the routine for spinal examination later in this chapter.

Biomechanical examination

As biomechanical abnormalities are one of the major causes of overuse injuries, it is essential to include this examination in the assessment of overuse injuries (Chapters 8 and 9). The biomechanical examination of the lower limb is illustrated in Chapter 8. The mechanics of the lower limb may be relevant to an upper limb injury (e.g. weak hip stabilisers resulting in overload of the elbow in sports involving repetitive throwing).

Functional testing

If a particular manoeuvre reproduces the patient's pain, then have the patient perform that manoeuvre in an attempt to understand why the pain has occurred. This can sometimes be done in the office (e.g. a deep squat) or it may be necessary to watch the athlete perform the activity at a training venue (e.g. a long jumper taking off or a gymnast performing a backward walkover). Video analysis may be helpful.

The examination routine

Inspection

It is important to observe the individual walking into the office or walking off the field of play as well as inspecting the injured area. Note any evidence of deformity, asymmetry, bruising, swelling, skin changes and muscle wasting. There may, however, be a degree of asymmetry due to one side being dominant, such as the racquet arm in a tennis player.

Range of motion testing (active)

Ask the athlete to perform active range of motion exercises without assistance. Look carefully for restriction of range of motion, the onset of pain at a particular point in the range, and the presence of abnormal patterns of movement. In many conditions, such as shoulder impingement or patellofemoral pain, the pattern of movement is critical to making a correct diagnosis.

If a patient's pain is not elicited on normal plane movement testing, examine 'combined movements' (i.e. movements in two or more planes). By combining movements and evaluating symptom response, additional information is gained to help predict the site of the lesion. Other movements, such as repeated, quick or sustained movements, may be required to elicit the patient's pain.

Range of motion testing (passive)

Passive range of motion testing is used to elicit joint and muscle stiffness. Injury may be the cause of joint stiffness. Alternatively, stiffness may already have been present and predispose to injury by placing excessive stress on other structures (e.g. a stiff ankle joint can predispose to Achilles tendinopathy). Range of motion testing should include all directions of movement appropriate to a particular joint, and should be compared both with normal range and the unaffected side. Overpressure may be used at the end of range to elicit the patient's symptoms.

Palpation

Palpation is a vital component of examination and precise knowledge of anatomy, especially surface anatomy, optimises its value. At times it is essential to determine the exact site of maximal tenderness (e.g. in differentiating between bony tenderness and ligament attachment tenderness after a sprained ankle). When palpating soft tissues, properties of the soft tissue that need to be assessed include:

- resistance
- muscle spasm
- tenderness.

Palpate carefully and try to visualise the structures being palpated. Commence with the skin, feeling for any changes in temperature or amount of sweating, infection or increased sympathetic activity. When palpating muscle, assess tone, focal areas of thickening or trigger points, muscle length and imbalance.

It is important not only to palpate the precise area of pain (e.g. the supraspinatus tendon attachment), but also the regions proximal and distal to the painful area, such as the muscle belly of the trapezius muscle. Determine whether tenderness is focal or diffuse. This may help differentiate between, for example, a stress fracture (focal tenderness) and periostitis (diffuse tenderness).

To palpate joints correctly, it is important to understand the two different types of movement present at a joint. Physiological movements are movements that patients can perform themselves. However, in order to achieve a full range of physiological movement, accessory movements are required. Accessory movements are the involuntary, interarticular movements, including glides, rotations and tilts that occur in both spinal and peripheral joints during normal physiological movements. Loss of these normal accessory movements may cause pain, altered range or abnormal quality of physiological joint movement. Palpation of the spinal and peripheral joints is based on these principles. An example of palpation of accessory movements involves posteroanterior pressure over the spinous process of the vertebra, producing a glide between that vertebra and the ones above and below.

Ligament testing

Ligaments are examined for laxity and pain. Specific tests have been devised for all the major ligaments of the body. These involve moving the joint to stress a particular ligament. This may cause pain or reveal laxity in the joint. Laxity is graded into +1 (mild), +2 (moderate) and +3 (severe). Pain on stressing the ligament is also significant and may indicate, in the absence of laxity, a mild injury or grade I ligament sprain. A number of different tests may assess a single ligament; for example, the anterior drawer, Lachman's and pivot shift tests all test ACL laxity.

Strength testing

Muscles or groups of muscles should be tested for strength and compared with the unaffected side. Muscle weakness may occur as a result of an injury (e.g. secondary to a chronic joint effusion) or may be a predisposing factor towards injury. Muscle strength can be assessed manually or with the use of a hand-held dynamometer.

Testing neural mechanosensitivity

Advances in the understanding of neural mechanosensitivity have improved our awareness of why pain occurs in chronic overuse injuries and pain syndromes. Changes in neural mechanosensitivity are an important component of these disorders; see Chapter 4 (overuse) and Chapters 5 and 6 (pain).

Just as restrictions of the normal mechanics of joints and muscles may contribute to symptoms, restriction of the normal mechanics of the nervous system may also produce pain. Certain movements require considerable variations in nerve length. Neurodynamic testing (formerly known as 'neural tension tests') examines restriction of these normal mechanics and their effect on the patient's symptoms. Treatment aims to restore normal nerve mechanics.

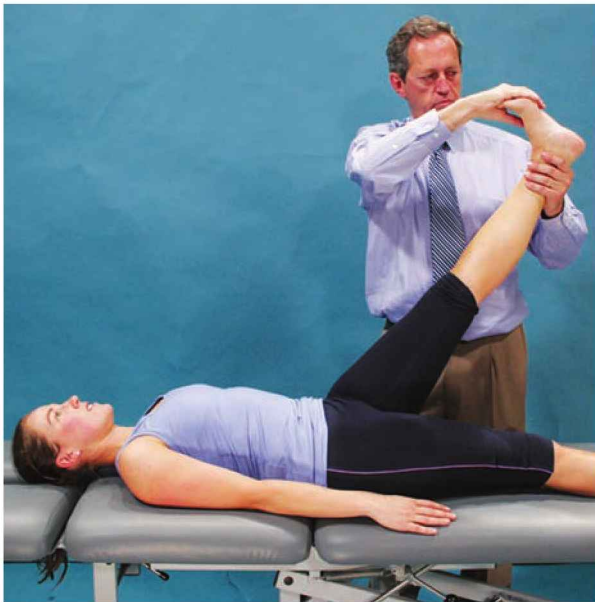
Neurodynamic tests use movement to systematically increase neural mechanosensitivity. The tests may provoke the presenting symptoms or, alternatively, other symptoms such as pins and needles or numbness. The amount of resistance encountered during the test is also significant, especially when compared with the uninjured side. The assessment of symptom production and resistance may be affected by each step in the neurodynamic test (see Figs 15.2 to 15.5). This may give an indication of the location of the abnormality.

The main neurodynamic tests are:

- straight-leg raise (SLR) (Fig. 15.2a and b)
- slump test (Fig. 15.3a-e)
- neural Thomas test (Fig. 15.4a and b)
- upper limb neurodynamic test (Fig. 15.5a-e).

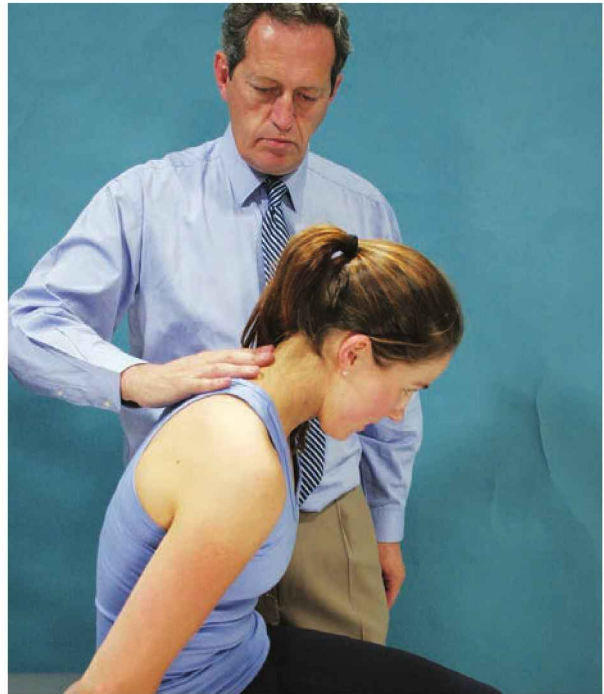


(a)

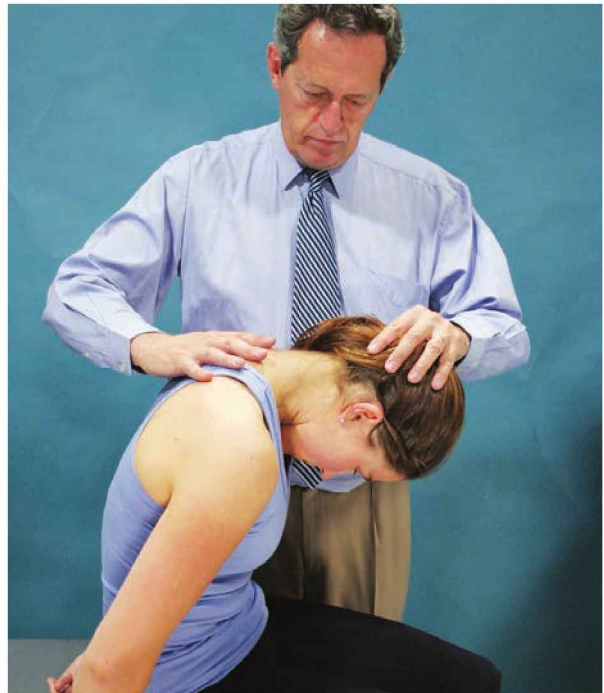


(b)

Figure 15.2 Straight leg raise (a) Patient lies supine. The examiner places one hand under the Achilles tendon and the other above the knee. The leg is lifted perpendicular to the bed with the hand above the knee, preventing any knee flexion (b) Dorsiflexion of the ankle is added. Eversion and toe extension may sensitise this test further. Other variations can be added (Table 15.1)



(a)

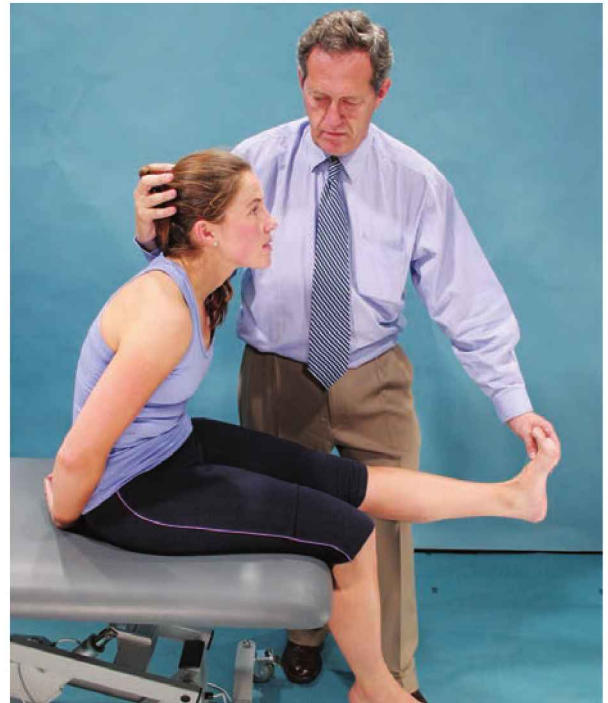


(b)

Figure 15.3 Slump test (a) Patient slumps forward and overpressure is applied. The sacrum should remain vertical (b) Patient is asked to put chin on chest and overpressure is applied

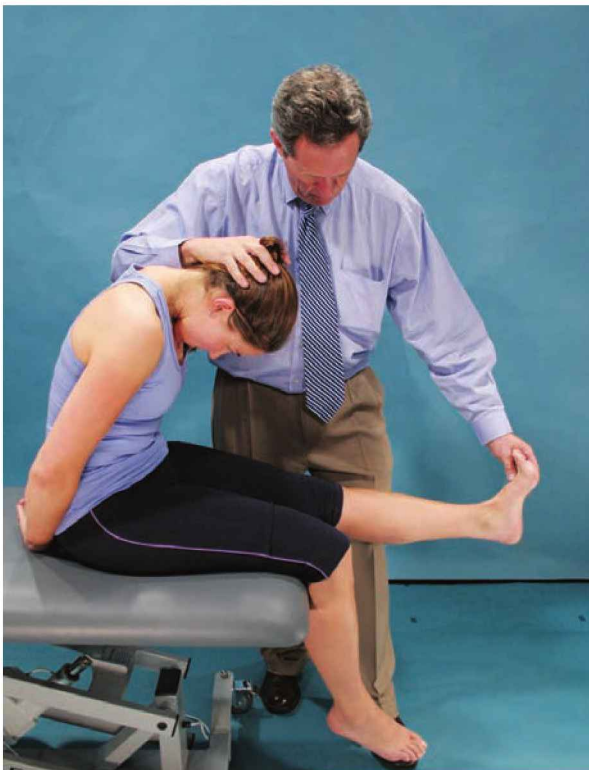


(c)



(e)

Figure 15.3 (cont.) (c) Patient actively extends one knee (d) Patient actively dorsiflexes the ankle and overpressure may be applied (e) Neck flexion is slowly released. Steps (d), (e) and (f) are repeated with the other knee. Other variations can be added (Table 15.1)

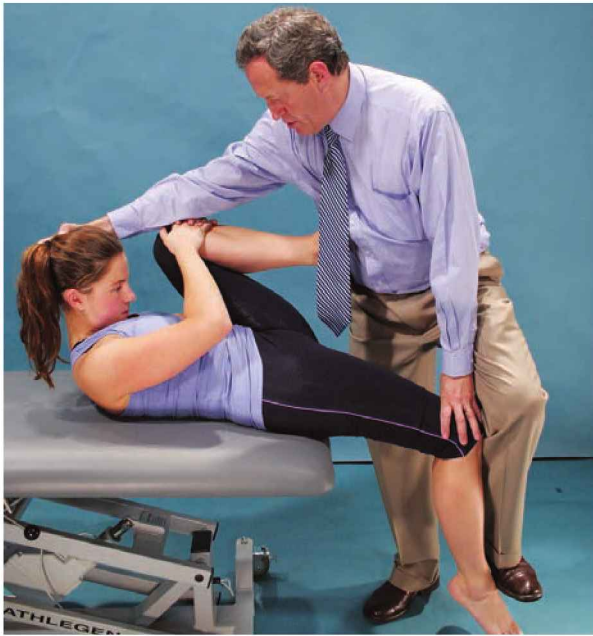


(d)



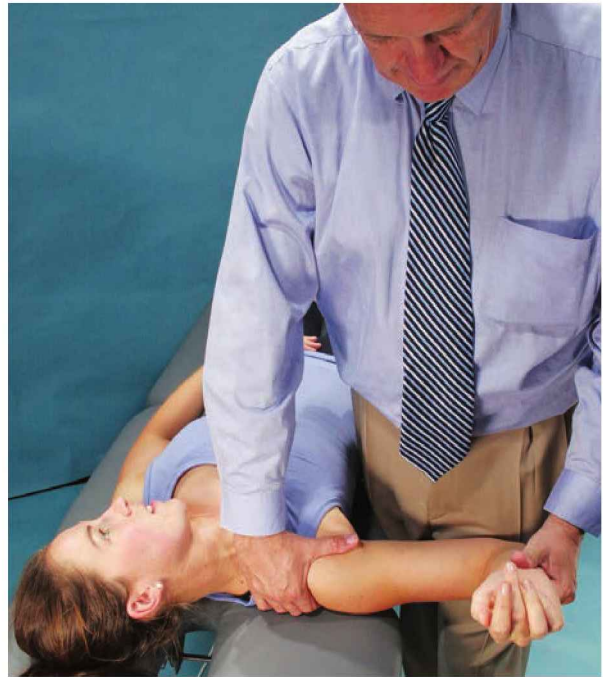
(a)

Figure 15.4 Neural Thomas test (a) Patient lies supine over the end of the couch in the Thomas position



(b)

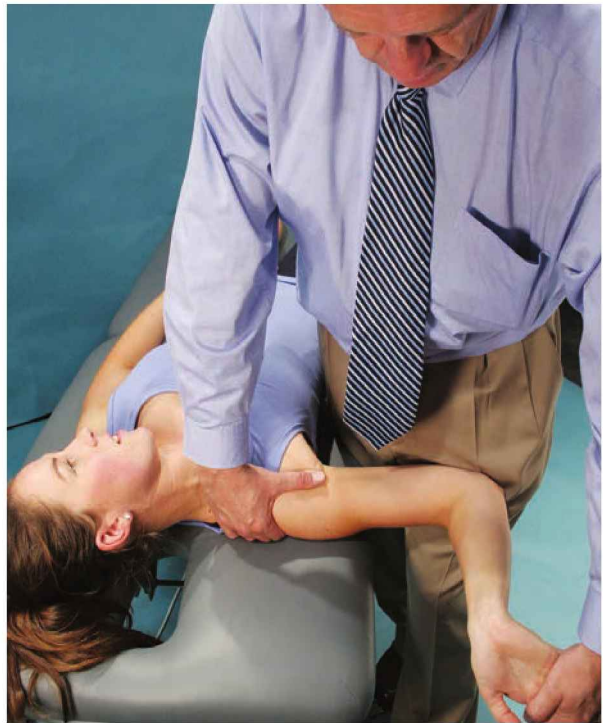
Figure 15.4 (cont.) (b) Patient's neck is passively flexed by the examiner, then the examiner passively flexes the patient's (right) knee with his leg



(b)

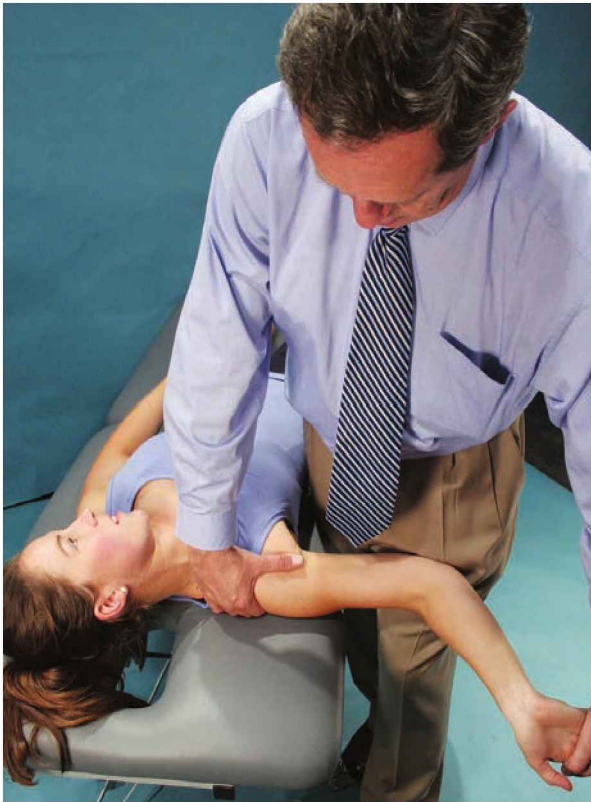


(a)

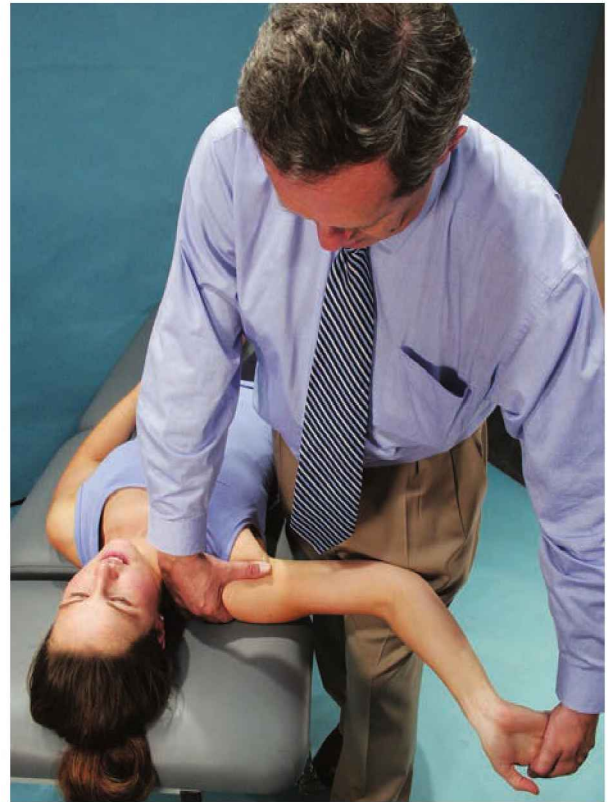


(c)

Figure 15.5 Upper limb neurodynamic test (a) Patient lies supine close to the edge of the couch. Neck is laterally flexed away from the side to be tested (b) The shoulder is depressed by the examiner's hand (left) and the arm abducted to approximately 110° and externally rotated (c) The forearm is supinated and the wrist and fingers extended



(d)



(e)

Figure 15.5 (cont.) (d) The elbow is extended to the point of onset of symptoms (e) The neck position returns to neutral and is then laterally flexed towards the side of the test. Any change in symptoms is noted. Other variations can be added (Table 15.1)

A summary of the tests, the methods, user guidance, normal responses and variations of each test is shown in Table 15.1.

A neurodynamic test can be considered positive if:

- it reproduces the patient's symptoms
- the test response can be altered by movements of different body parts that alter the neural mechanosensitivity
- differences in the test occur from side to side and from what is considered normal.

Neurodynamic tests are non-specific but form an extremely useful part of the examination. Abnormalities of neural mechanosensitivity should lead the clinician to examine possible sites of abnormality, especially the spine. The techniques used in neurodynamic tests can also be used as a treatment procedure. This is discussed in Chapter 17.

Spinal examination

Clinical experience suggests that spinal abnormality (e.g. hypomobility) can present in various ways. The

presentation may be as pain or injury, and this may occur either locally (at the spine) or distantly. Examples for both upper limb and lower limb spinal abnormalities are given in Table 15.2. The pathophysiology underlying these concepts has been discussed in Chapters 5 and 6.

In patients presenting with upper limb pain, examine the cervical and upper thoracic spines. Examine the lumbar spine (including the thoracolumbar junction) in any patient presenting with lower limb pain. An abnormal neurodynamic test strongly indicates a spinal component to the pain. However, a negative neurodynamic test does not rule out the possibility of a spinal component.

Begin examining the relevant area of the spine by assessing range of movement with the patient standing. The patient should then lie prone on a firm examination table so the examiner can palpate the vertebrae centrally over the spinous processes and laterally over the apophyseal joints to detect any hypomobility and/or tenderness. Hypomobility or tenderness at a level appropriate to that of the patient's symptoms indicate the site is a possible source of referred pain (Chapters 5 and 6).

Table 15.1 Neurodynamic tests

Test	Method	Indications	Normal response	Variations
Straight leg raise (Fig. 15.2)	Patient lies supine Leg extended Clinician lifts leg	Leg pain Back pain Headache	Tightness and/or pain in posterior knee, thigh and calf	Ankle dorsiflexion Ankle plantarflexion/ inversion Hip adduction Hip medial rotation Passive neck flexion
Slump test (Fig. 15.3)	Patient sitting Slumps Neck flexion Knee extension Ankle dorsiflexion Release neck flexion	Back pain Buttock pain Leg pain	Upper thoracic pain Posterior knee pain Hamstring pain	Leg abduction (obturator nerve) Hip adduction Hip medial rotation Ankle and foot alterations
Neural Thomas test (Fig. 15.4)	Patient lies supine Hip extension Neck flexion Knee flexion	Groin pain Anterior thigh pain	Quadriceps pain and/or tightness	Hip abduction/adduction Hip medial/lateral rotation
Upper limb neurodynamics test (Fig. 15.5)	Patient supine towards side of couch Cervical contralateral flexion Shoulder girdle depression Shoulder abducted to 110° and externally rotated Forearm supination Wrist/fingers extended Elbow extended	Arm pain Neck/upper thoracic pain Headache	Ache in cubital fossa Tingling in thumb and fingers	Forearm pronation Wrist deviation Shoulder flexion/extension Add straight leg raise

Table 15.2 Examples of how spinal abnormality can manifest locally or distantly, with either pain or injury in the upper limb and lower limb

Presentation	Local manifestation	Distant manifestation
Upper limb		
Pain	Hypomobility of C5–C6 joint presenting as neck pain	Hypomobility of C5–C6 joint presents as elbow pain
Injury		Hypomobility of C5–C6 joint predisposing to lateral elbow tendinopathy in a tennis player
Lower limb		
Pain	Hypomobility of L5–S1 joint presenting as lumbosacral pain	Hypomobility of L5–S1 joint presents as buttock and hamstring pain
Injury		Hypomobility of L5–S1 joint predisposing to a hamstring tear in a sprinter

After detecting spinal abnormality on examination, perform a trial treatment (Chapter 17) and then reassess the patient's symptoms and signs. If there is a change in the pain and/or range of movement, then this strongly suggests that the spine is contributing to the symptoms.

Occasionally, palpation of a particular site in the spine will actually reproduce the patient's symptoms distant from the spine. It is important to understand, however, that even if the symptoms are not produced by palpation of the spine, this does not rule out the possibility of a spinal component.

Biomechanical examination

The role of abnormal biomechanics in the production of injuries, especially overuse injuries, is discussed in Chapters 8 and 9. Because abnormal biomechanics can contribute to any overuse injury, all clinicians need to perform a biomechanical examination. As with other components of the examination, it is important to develop a routine for the assessment of biomechanical abnormalities. A routine for the assessment of lower limb biomechanics is illustrated in Chapter 8.

Technique

Faulty technique is another common cause of injury. Technique faults associated with particular injuries are discussed in Chapter 9. While the clinician cannot be aware of all techniques in various sports, he or she should be able to identify the common technique faults in popular activities (e.g. pelvic instability while running or faulty backhand drive in tennis). Clinicians should seek biomechanical advice and assistance with assessment from the athlete's coach or a colleague with expertise in the particular area. Video analysis with slow motion or freeze frame may be helpful.

Equipment

Inappropriate equipment predisposes to injury (Chapter 12). Inspect the athlete's equipment, such as running shoes, football boots, tennis racquet, bicycle or helmet.

Differential diagnosis

At the conclusion of the examination, consider the differential diagnosis and possible predisposing factors. If the clinician is certain of the diagnosis and of the predisposing factors, then counselling and treatment can begin. However, in many cases, further information may be required and the clinician must decide what, if any, investigations may be needed.

DIAGNOSTIC IMAGING

This section introduces imaging—which has provided dramatic advances for sports medicine, and is widely used

in clinical medicine. As well as providing remarkable insight to sports medicine pathologies, superior imaging capability has brought with it new challenges for clinicians: 'Yes, that bright signal is abnormal most of the time but what does that mean in this patient?'. But before we tackle that difficult question, we begin with what we know to be true—the five habits of highly effective clinicians, with respect to imaging. Every top sports clinician is highly skilled at interpreting sports imaging and he or she works closely with top musculoskeletal radiologists to perform at the level that champions expect, and every patient who sees a specialist deserves.⁶

The five imaging-related habits of highly effective sports medicine clinicians

In certain clinical settings, appropriate investigations can confirm or exclude a diagnosis suggested by the history and physical examination. A conventional radiograph can confirm that a cyclist has fractured her clavicle. Computed tomography (CT) with multiplanar reformat and three-dimensional volumetric rendering can provide the precise configuration of that fracture. MRI can rule out a stress fracture in a dancer's foot if there is no sign of any abnormal signal.

PRACTICE PEARL

In many clinical contexts imaging does not identify a single structure that is the source of pain. Imaging the patient with back pain is the historical example used to illustrate that point but there are many more examples.

As an example, MRI of a professional baseball player's painful shoulder or elbow illustrates the point clearly.⁷⁻⁹ Most baseball players have structural changes visible on MRI of their shoulder and elbow (e.g. thickening of their ulnar collateral ligament); one needs to consider if this appearance is 'normal' for that player or not. Other very prominent regions where structural changes are often evident on MRI but which may or may not be symptomatic include the tennis player's elbow, the football player's ankle, the fast bowler's back in cricket. The take-home message is that imaging alone cannot substitute for careful history taking and a comprehensive examination (Chapter 14). Because diagnosis is complex, the leading clinicians in our field have embraced the following behaviours.

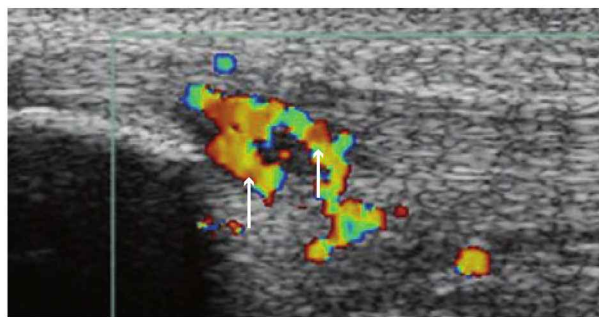
1. Understand imaging results

The sport and exercise medicine physician must be able to interpret investigation results him or herself. Learn imaging anatomy of the musculoskeletal system—it is only a fraction of what radiologists have to know.¹⁰ It is unwise to rely blindly on imaging reports. A competent sports clinician knows that about 25% of asymptomatic elite jumping athletes have ultrasound appearances of structural change in their patellar tendons (hypoechoic regions, Fig. 15.6a and b).^{11, 12} This is critical knowledge as the patient may have no pain there; in fact, the extremity may have been scanned as the asymptomatic contralateral knee for comparison with the symptomatic side. Without requisite clinical awareness, the clinician could let the test guide him or her to a potentially 'false positive' result (depending on the criteria for 'positive', see below). Management must be guided by clinical assessment, not by imaging appearance.

- *Timing of imaging tests.* Imaging may be performed too early to detect a pathology. That radiography can



(a)



(b)

Figure 15.6 Variations from 'normal' imaging are common among athletes who place great loads on tissue. (a) Grey-scale and (b) Doppler ultrasound appearances of the patellar tendon in a volleyball player, demonstrating hypoechoic regions (arrows in a) and hyperaemia (arrows in b). Clinical reasoning needs to consider the high prevalence of these findings in asymptomatic players¹²

be performed too early to detect a stress fracture is widely known and therefore a 'negative' result does not exclude this diagnosis (high or moderate negative likelihood ratio, so not good for ruling out that pathology).

- *A 'true' finding may be age-related.* For example, triangular fibrocartilage tears of the wrist seen on MRI can be a normal finding, even in a non-athlete, for patients over 50 years of age. Clinical correlation with the patient's symptoms and the physical examination is crucial.
- *The wrong test can give false positive or false negative information.* For example, a routine MRI of the shoulder in a 26-year-old baseball pitcher will miss most labral tears; an MR arthrogram is needed to make the diagnosis. Radiologists are not only 'image interpreters', they are also consultants who can advise on the choice of imaging tests to best demonstrate pathology. Indeed, the most accurate imaging test for a given musculoskeletal pathology may not necessarily be the most sophisticated or complicated test.
- *Be able to judge the adequacy of the quality of obtained images for diagnostic purposes.* Be alert so as not to draw conclusions from imaging tests that may be incomplete or are limited by image degradations. Also, one needs to understand that each imaging test may be specifically obtained to evaluate for one type of pathology but not others. For example, two-view radiographs of the shoulder in internal and external rotation to evaluate for calcific tendinopathy should not be used to exclude glenohumeral dislocation.
- *Intravenous contrast is usually reserved for CT or MRI cases in which the diagnosis of neoplasm or tumour is being considered.* It is very important to indicate as such on the requisition if these need to be ruled out.
- *It is important to minimise ionising radiation, especially in children and adults aged under 30 years.* The dose is especially significant in nuclear medicine bone scans, positron emission tomography (PET) scans and CT scans. If any of these tests are truly indicated, the principle of ALARA (as low as reasonably achievable) with respect to dose should be respected by the imaging department. Note that there is no ionising radiation in MRI, and the dose is negligible in routine conventional radiographs.

Why do we share these detailed examples about clinicians needing to understand tests? Because the days of thinking that 'imaging reveals X, therefore treatment is Y' are essentially over. Imaging fits into the treatment algorithm and helps update the clinician's thinking of the likelihood of a diagnosis (See Likelihood ratios, Chapter 14). The clinician must understand imaging because it is a tool, not a solution, and far from a panacea.

2. Only order imaging that will influence management

It is inappropriate to perform imaging to confirm an already obvious diagnosis, or a diagnosis for which the investigation result will not influence management. Most concussions are diagnosed clinically and do not need brain CT or MRI. Most cases of back pain in sport are not helped by adding low back imaging (radiography, CT or MRI). If an ankle sprain does not appear to have caused a fracture (and there are clinical guidelines to help with the assessment), initial management does not include any imaging. Players/athletes can put clinicians under pressure to order tests; in theory this should always be resisted. In real life, it can be difficult to resist such pressure in the professional sport setting so we will not pretend that ideal management is always followed. But understanding the usefulness of tests and test metrics (Chapter 14) as well as normal patterns of imaging in sport (principle 1, above), allows the clinician to have a valuable conversation with the patient/athlete, and the patient to make an informed decision (see also Figure 2.1 in Chapter 2).

3. Explain the imaging to the patient

Give the patient an understanding of the rationale behind each imaging test. An athlete who complains of persistent ankle pain and swelling several months after an ankle sprain may need a radiograph and MRI. If the patient is merely told that radiography is necessary to exclude bony damage, he or she might become confused when told that the radiograph is normal, but that further investigations are required to exclude bony or osteochondral damage. Also, be sure to alert patients undergoing a procedure that involves contrast (e.g. MR arthrogram) that there will be an injection. It is helpful to provide the patient with resources (either as printed material or as weblinks to videos) explaining the investigation. Follow-up plans should be made explicit in the clinical consultation.

4. Provide relevant clinical findings on the requisition

Accurate and complete clinical information on requisition forms helps to avoid imaging and reporting errors. When particular radiographic views are required, they should be specified. If you cannot remember the names of certain views, write that down on the request forms—the radiographer will generally know them and, if not, the radiologist will. If there is uncertainty, it is always helpful to call the radiologist in advance to discuss the best way to image a patient. Do not assume that radiographers are familiar with the specific views required for sports-related musculoskeletal conditions (e.g. correct anteroposterior [AP], ‘frog’ and Dunn views for hip femoroacetabular impingement [FAI], skyline views for patellofemoral joint evaluation, the Bernageau view

for shoulder instability). Remember that weight-bearing views are important to assess suspected osteoarthritis at the hip, knee and ankle. ‘Functional’ views (with the patient placing the joint in the position of pain) are useful for anterior and posterior impingement of the ankle (Chapter 42).

5. Develop a close working relationship with investigators

Optimising communication between colleagues improves the quality of the service.⁶ Discuss optimal imaging sequences with radiographers, and view the images together with radiologists providing clinical input. Regular clinical-radiological rounds or case presentations should be encouraged. Digital imaging and telemedicine has made this much easier.

SPECIFIC IMAGING MODALITIES

In this overview of imaging modalities commonly used in sports medicine, our goal is to highlight some key principles that junior clinicians have told us they would have liked to have known earlier in their careers. Read this together with other resources such as the comprehensive, authoritative and beautifully illustrated text in this field: *Imaging in Sports-Specific Musculoskeletal Injuries* (Fig. 15.7).¹⁰

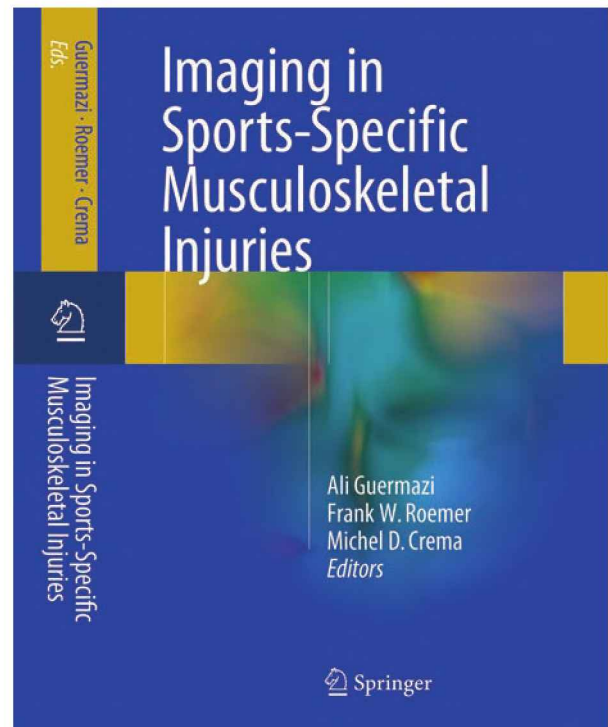


Figure 15.7 A great resource for specialists who work in sports medicine.

REPRODUCED WITH PERMISSION FROM GUERMAZI A, ROEMER F, CREMA M. *IMAGING IN SPORTS-SPECIFIC MUSCULOSKELETAL INJURIES*. SPRINGER, 2015.



Figure 15.8 Plain radiograph showing an osteochondral fracture of the lateral talar dome COURTESY OF IF ANDERSON

Conventional radiography

Even as we approach 2020, radiography still provides diagnostic information about bony abnormalities, such as fractures, dislocations, dysplasia and calcification (Fig. 15.8), and is highly cost-effective. Correctly positioning the patient is vital for obtaining radiographic images that are meaningfully interpretable. A minimum of two orthogonal views is required to evaluate any bone adequately. Complex joints such as the ankle, wrist or elbow may require additional or specialised views. Weight-bearing or 'stress' views may give further information for weight-bearing structures and joints.

MRI: massive blessing for active patients

The 2003 Nobel Prize was awarded to the inventors of MRI. This imaging method has revolutionised sports medicine by providing clinicians with remarkably detailed information about structure, especially soft tissues, and helpful information about dynamic pathophysiology. Because of MRI's unparalleled contrast resolution, commonly injured musculoskeletal tissues such as the menisci in the knee, the labrum in the shoulder and hip, spinal discs and joint surfaces can be visualised non-invasively and without ionising radiation.

MRI allows clinicians to appreciate fairly dynamic processes such as bone and cartilage turnover, changes in tendon structure even in the absence of rupture, and recovery of ligament after acute sprain.

The physics that underpins MRI is well beyond the scope of this book. At a most superficial level, MRI can be understood to rely on the biology that free water protons exist within a tissue sample. Those protons align with the external magnetic field of the magnet. When a series of radiofrequency (RF) pulses are applied to the tissue sample in the magnet, protons change their alignment relative to the external magnetic field and release energy which creates the MR image.

Unparalleled sensitivity to detect deviation in structure from anatomical norms does not, however, equal unparalleled accuracy in diagnosis for the patient (Chapter 14). Erroneous interpretation of an MRI can have serious clinical consequences. For example, if a 50-year-old person's knee MRI is misinterpreted, a patient who in reality has patellofemoral pain may be inappropriately slated for arthroscopic meniscal surgery (Fig. 15.9).

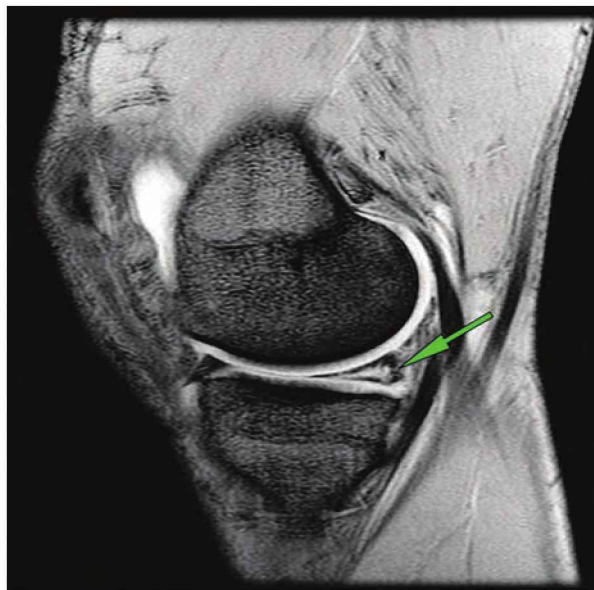


Figure 15.9 Incidental MRI-detected meniscal pathology occur in 19% of women aged 50–59 years¹³. The figure shows a horizontal cleavage tear of the posterior horn of the medial meniscus (arrow). Whether or not this MRI-detected lesion is clinically relevant in a patient presenting with knee pain depends on the entire clinical picture, including detailed history and physical examination findings. The take-home message is that a clinician cannot, and should not, make a diagnosis and decide how to treat the patient solely based on MRI findings

What does this MRI tell me? Tips for clinicians

The four most common sequences a clinician will see are outlined below and shown in Figure 15.10.

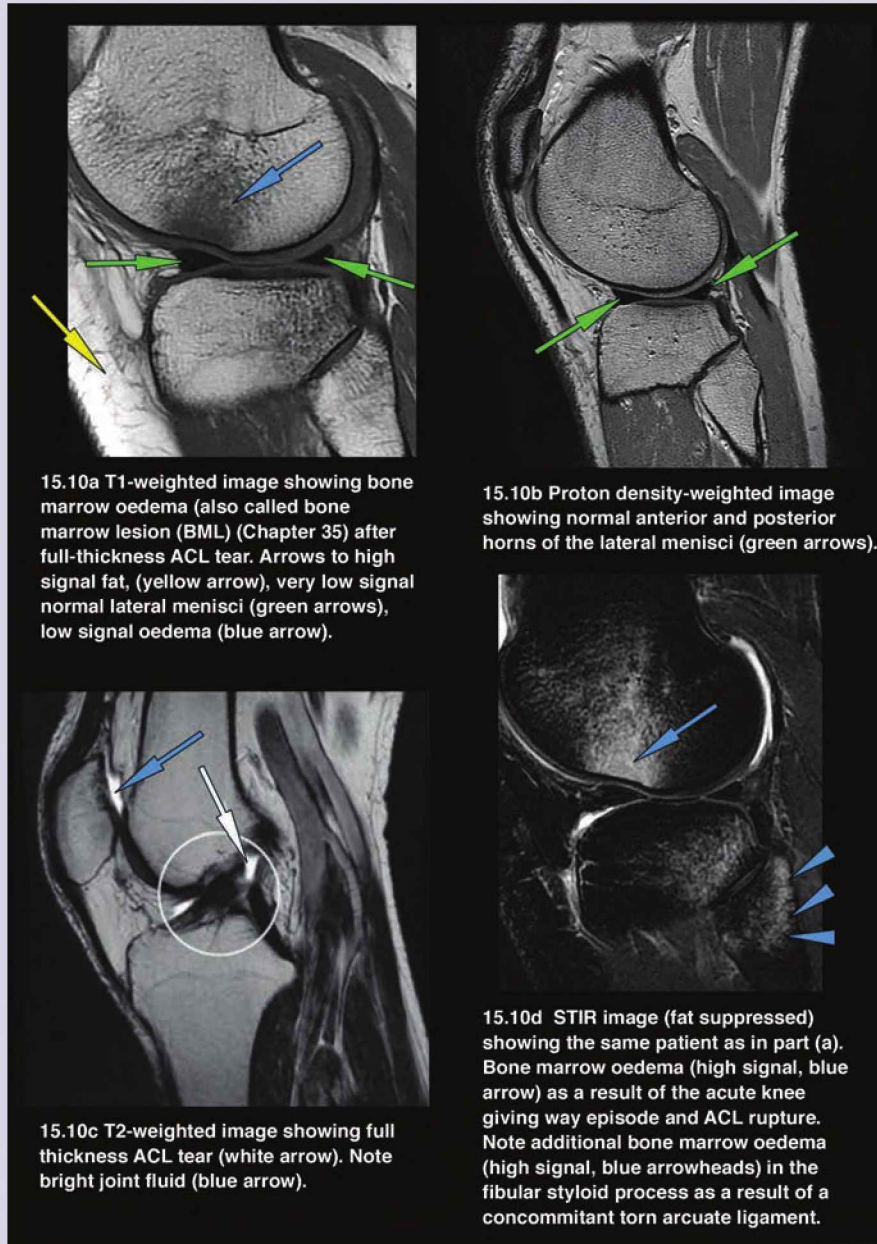


Figure 15.10 The legend for the arrows/arrowheads to structures in the four MRI sequences is as follows: fat = yellow; tendon and ligament = off-white; water = blue; menisci = green

- T1-weighted** provides sharp anatomical detail, shows bone marrow and is good for meniscal pathology. It lacks the sensitivity to detect soft tissue injury (Fig. 15.10a). MRI signal key: fat = bright; muscle = intermediate; water, tendons and fibrocartilage all dark.
- Proton density-weighted** is good for imaging menisci and ligaments (Fig. 15.10b). MRI signal key: fat = bright or intermediate signal, calcium, tendons and fibrocartilage, all dark; water = intermediate.
- T2-weighted** is highly sensitive for most soft tissue injury, especially tendons. Abnormal tendons have high signal intensity (bright) which contrasts with normal tendons black (arrowhead) (Fig. 15.10c). MRI signal key: water = bright; fat = intermediate muscle, hyaline and fibrocartilage all dark.
- STIR** highlights excess water (blue arrow and blue arrowheads) which can occur due to bone stress and bone marrow oedema (shown), joint fluid and soft tissue pathology. This is the sequence of choice for bone stress injuries or subtle, radiographically occult fractures (Fig. 15.10d). MRI signal key: water = very bright; fat, muscle, menisci all dark.

The price of additional medical data is that the clinician who cares for the patient needs to determine how heavily to weigh the imaging information. Specifically, the clinician must arrive at a post-test probability of a diagnosis based on how much the MRI result (positive or negative) changes the pre-test probability of the diagnosis. (See Fagan's nomogram, Chapter 14.)

Specific features and patient benefits of MRI

MRI is now a routine part of the medical landscape the world over, although access to MRI for sports medicine imaging is limited in some regions. The tide is heading powerfully in one direction towards greater and greater utilisation of MRI in sports medicine. Paradoxically, in developed countries that have private medicine (radiology), too much imaging and overdiagnosis, associated with overtreatment (particularly arthroscopic surgery) has a deleterious effect on health. So the challenge for both policy makers and skilled clinicians is to choose imaging wisely. Towards that aim, we first review the features of MRI as a tool for clinicians working in sport and promoting physical activity.

MRI has the following features which translate into patient and clinician benefits.

- MRI is non-invasive, so it has dramatically reduced the need for diagnostic arthroscopy.
- It can provide images in multiple planes. Need a sagittal view of the knee to image the patellar tendon? Sure! How about a coronal to assess joint surfaces and menisci? No problem!
- MRI does not involve ionising radiation. This is particularly important in the age group often treated by clinicians in sports medicine: adolescents, as well as for professional athletes who may have many clinical encounters and thus need many scans.
- MRI has superb contrast resolution for soft tissues, allowing accurate assessment of most ligaments, tendons and hyaline and fibrocartilage. Note that although bones are also well-assessed, if the clinical concern is one of primarily bony pathology (e.g. lumbar spondylolysis), CT may be more appropriate, bearing in mind radiation dose.
- It is ideal to assess patients who have had soft tissue trauma (e.g. contact injuries) and it complements the use of CT in high-velocity injuries.
- MRI has an emerging role in helping clinicians evaluate overuse injuries. Anatomy in patients in the sports medicine setting is markedly affected by microtrauma/adaptation to load and this is increasingly being well characterised. In the late 1980s, journals commonly carried treatment recommendations from 'experts' based on false premises of what was 'normal' in athletes.¹⁴ The problem was that 'normal' for an elite athlete had

not been well documented. The subspecialty of sports radiology and greater experience of musculoskeletal radiologists with sport is addressing that challenge.⁶ As with all research, knowledge translation remains a challenge; getting the key messages to the community of clinicians who will apply it.

Difference added by MRI sequences

Spin echo (SE), gradient echo (GRE) and inversion recovery (IR) sequences are the basic categories of sequences used in musculoskeletal imaging.³ T1-weighted, T2-weighted, proton density-weighted and short tau inversion recovery (STIR) sequences are often utilised (Figure 15.10). Fat-suppressed fast spin echo sequences are used in almost every musculoskeletal protocol to demonstrate bone marrow oedema, focal cartilage defects and tendon/ligament pathology with the highest conspicuity.

Other clinical points about MRI

The clinical workhorse for sports MRI is the 1.5 T (Tesla) MRI platform. For almost all clinical indications, MRI at this field strength is generally sufficient. The 3 T magnets, which are more expensive to purchase, offer higher spatial resolution, and therefore may offer advantages in evaluating small structures, such as tendon pulley injuries in rock climbers (although there are currently no trials to support this). Advanced applications in 3 T may prove useful in the future in dedicated cartilage imaging.

The contrast agent for MRI is gadolinium, which has an excellent safety profile, even more so than the safe iodinated contrast agents used in CT. It is used intravenously for musculoskeletal imaging in cases in which tumour or neoplasm are being considered, and in an intra-articular fashion to optimally evaluate hyaline cartilage, such as shoulder and hip labra.

A less common architecture for MRI is so-called 'open' units, which do offer advantages in allowing patients with claustrophobia to be scanned; this occurs at the expense of image quality. The weight-bearing and kinematic options that these units provide are still investigational in nature (Fig. 15.11).

Physiologic (also called compositional) imaging of cartilage can be performed at 1.5 T and 3 T, but not accurately at lower field strengths. Special sequences can assess for loss of cartilage proteoglycan content (T1-rho mapping and dGEMRIC) and for collagen degradation (T2 mapping). These techniques hold great promise in biochemical, pre-morphologic chondral assessment in early osteoarthritis, and for follow-up of interventions (e.g. chondral surgery and, one day, biologics that can block the progression of osteoarthritis). However, currently their clinical role is undefined.



Figure 15.11 The 'open' MRI instrument allows the patient to be weight-bearing. In this figure the model is wearing a knee coil

IMAGE COURTESY OF CENTRE FOR HIP HEALTH AND MOBILITY, UNIVERSITY OF BRITISH COLUMBIA [HTTP://WWW.HIPHEALTH.CA/](http://www.hiphealth.ca/)

There are a few strict contraindications to MRI (e.g. certain brain aneurysm clips, neurostimulators and cardiac pacemakers) but, contrary to popular medical opinion, patients with metallic orthopaedic hardware and metallic surgical clips outside the brain, in place for more than 6 weeks, can be safely scanned.

MRI can be overly sensitive to abnormal tissue signals and, thus, provide false positive results. In asymptomatic athletes in numerous studies, MRI revealed structural appearances consistent with significant injury but the athletes were asymptomatic and high functioning.^{12, 13} This emphasises the need for the appropriate selection of patients for investigation and careful clinical-imaging correlation.¹ As with any medical investigation, diagnostic errors can occur; ideally, images should be read by an experienced musculoskeletal MRI radiologist.

Ultrasound scan (for diagnosis)

High-resolution ultrasound scanning (Fig. 15.12) with 10–12 MHz probes is a painless method of imaging tendons, muscles and other soft tissues without exposing the patient to any radiation.¹⁵ It allows clinical correlation—imaging with the patient reproducing his or her pain and providing feedback to the radiologist. Adding to its diagnostic utility, both sides can be readily imaged for comparison. Other advantages include its dynamic nature (the patient can move the part and/or reproduce pain during the examination), short examination time and ability to guide therapeutic aspiration or injection under real time.

Ultrasound is the imaging instrument of choice 'in clinic'.^{16, 17} As it is an excellent way to visualise tendon and muscle, it can be the workhorse in a sports clinician's office where patients present with injuries of these tissues.



Figure 15.12 High-resolution ultrasound scanning with 10–12 megahertz (MHz) probes is a painless method of imaging tendons, muscles and other soft tissues without exposing the patient to any radiation

Units are priced to be affordable in the clinic because the unit cost, installation and service are of the order of 100 times cheaper than MRI.

PRACTICE PEARL

Ultrasound has been referred to as the sports physician's stethoscope and the analogy applies to the portability of ultrasound—small units can be used on the sideline.¹⁸ Laptop-sized portable units have a different level of image quality to the larger units that are designed to stay in the clinic.

Clinicians in sport might consider ultrasound equipment as having three different levels.

1. The most expensive, best quality equipment would be located at a specialist radiology clinic. The radiologist and his/her team with the full battery of complementary imaging modalities provides the tertiary resource as well as continuing professional education (formal and informal) and research partnerships as appropriate.
2. Leading specialised clinical practices often choose to have quality ultrasound imaging capacity at the clinic to assist with speed of diagnosis, for injections with ultrasound imaging and because having ultrasound skills advances those clinicians understanding of anatomy.
3. Portable sideline equipment (laptop scanner shown in Fig. 15.13) still provides quality images but the users appreciate the limitation compared with that outlined in point 1.

Disadvantages of ultrasound include the less graphic images, the fact that it is more operator-dependent with



Figure 15.13 Portable sideline ultrasound unit
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respect to image quality than any other modality, and the fact that it cannot penetrate tissues to show deeper structures, such as shoulder/hip labra or anterior cruciate ligaments/menisci. The most commonly examined areas are large tendons, (e.g. the Achilles, patellar and rotator cuff tendons) and the muscles of the thigh and calf. Ultrasound can also demonstrate muscle tear, haematoma formation or early calcification, and may be useful in localising foreign bodies.

Ultrasound scanning is able to distinguish complete tendon rupture from other tendon abnormalities (e.g. tendinopathy). As with MRI, ultrasound imaging of elite athletes reveals morphological 'abnormalities' that are not symptomatic and do not appear to predict imminent tendon pain.

Real-time ultrasound examination during active movement (dynamic ultrasound) is particularly helpful in the evaluation of shoulder impingement. The use of ultrasound to help guide injection is discussed in Chapter 17.

The role of colour Doppler

Colour Doppler ultrasound gained popularity in sports medicine for the assessment of tendons in the early 2000s, as it was hoped that abnormal tendon flow detected using

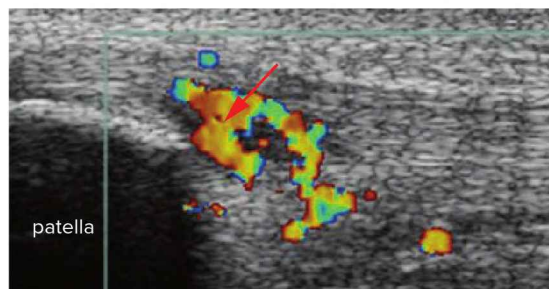


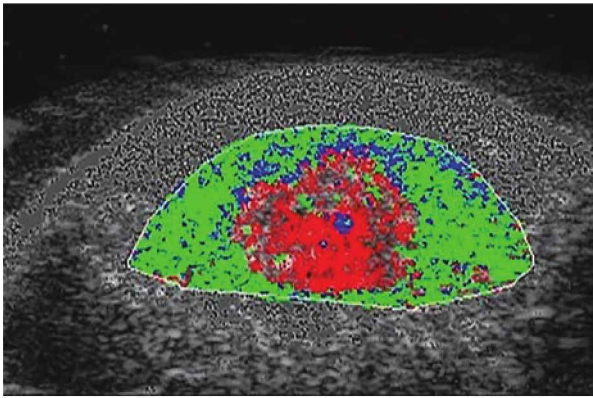
Figure 15.14 Sagittal colour Doppler ultrasound scan showing abnormal blood flow (arrow) at the mid- and deep surface of the patellar tendon near the patellar attachment

the colour Doppler feature would provide a better guide as to whether or not tendons were painful (Fig. 15.14). Longitudinal studies failed to show that colour Doppler ultrasound findings of vascularity predicted changes in symptoms.¹⁹ Also, exercise affects the level of vascularity, which adds a challenge to the clinical utility of this method. This modality is proving useful in certain areas of medicine but has dropped off the 'hot topics' list in sports medicine. To return to a theme from early in this chapter, there appears to be no substitute for careful clinical examination and patient monitoring to track patients' current progress and to predict outcomes.

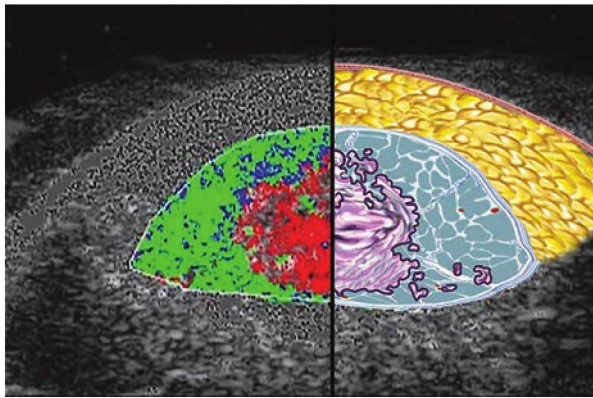
The role of ultrasound tissue characterisation

In the 2010s, a novel ultrasound modality, called ultrasound tissue characterisation (UTC), was introduced to evaluate human tendons (Fig. 15.15).²⁰ The innovation aims to provide a three-dimensional image of the tendon where structure is quantified (in contrast to conventional greyscale ultrasound which provides no numerical data). Data are obtained from the stability of pixel attributes (brightness) over a set scan length to a maximum of 25 scan slices at present. UTC has high reproducibility with excellent intra-observer and interobserver reliability.²⁰ Detailed descriptions of the use of this emerging technology are reported in various clinical research reports.^{20, 21}

Is this method ready for general use in 2017? No. Ultrasound tissue characterisation is currently a research instrument with potential that marketers would say is at the early stage of the 'product life cycle'.²² As with any new technology, there will be extreme claims as to its potential capacity; over time a body of evidence will emerge as to the best ways for clinicians and patients to benefit from the technological advance.²¹ At the time of writing, there is only one manufacturer of this patented technology (UTC imaging, Stein, The Netherlands) and the cost of an instrument is in the range of US\$40 000 to US\$60 000, depending on service costs and the options that the user chooses for elements such as probes.



(a)



(b)

Figure 15.15 Ultrasound tissue characterisation (UTC) of the Achilles tendon. Standard image (axial plane) in (a), and put in context in (b) with yellow being subcutaneous tissue. The tendon tissue would all be green in a normal tendon. This tendon has collagen pathology which is seen as blue, red and black signal. Green = echo-type I (normal tendon); blue = echo-type II; red = echo-type III; black = echo-type IV. Echo-types II–IV are abnormal

CT scanning

CT scanning (Fig. 15.16) allows cross-sectional imaging of soft tissue, calcific deposits and bone. CT scanning is particularly useful in evaluation of the spine, fractures in small bones and fractures in anatomically complex regions, such as the ankle, foot or pelvis. CT scanners provide high-resolution reconstructions of the imaging

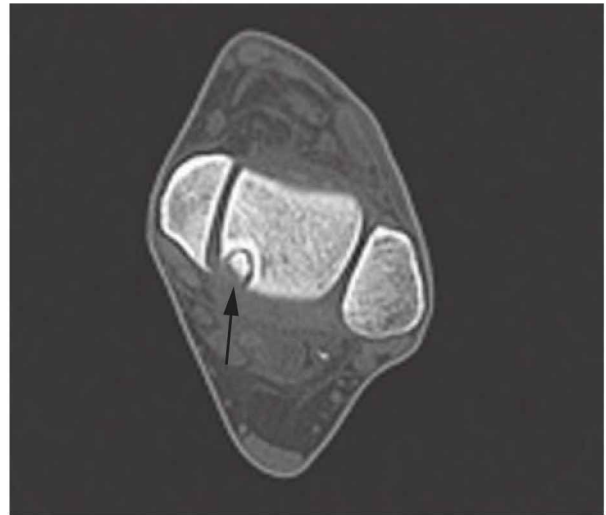


Figure 15.16 Axial CT scan showing cross-section of osteochondral lesion of the posteromedial talar dome (arrow)

data in any plane. Remember that CT defines bone details and can detect small areas of calcification better than MRI can, even though the latter will reveal occult bony abnormalities. Advances in multidetector CT technology allow rapid image acquisition and superb multiplanar reconstruction. The disadvantage of CT scanning is the significant radiation dose, especially in children.

Radioisotopic bone scan

Radioisotopic bone scan (scintigraphy) has fallen out of favour in sports medicine practice because of the availability of MRI with appropriate pulse sequences (e.g. STIR) which can often provide similar information about bone while adding anatomical localisation that was limited with bone scan. The high sensitivity and low negative likelihood if negative for stress fracture meant that the test was useful for ruling out stress fractures. Does radioisotopic bone scanning still have a role? Very rarely. Expert and experienced radiologists may use it at times to seek a tibial or a metatarsal stress fracture but it has essentially been superseded by MRI and CT.

Single photon emission computed tomography (SPECT) techniques are also used in sports medicine, particularly in the detection of stress fractures of the pars interarticularis of the lumbar spine, although the correct MRI sequence should be able to detect these changes.

REFERENCES

References for this chapter can be found at www.mhhe.com/au/CSM5e